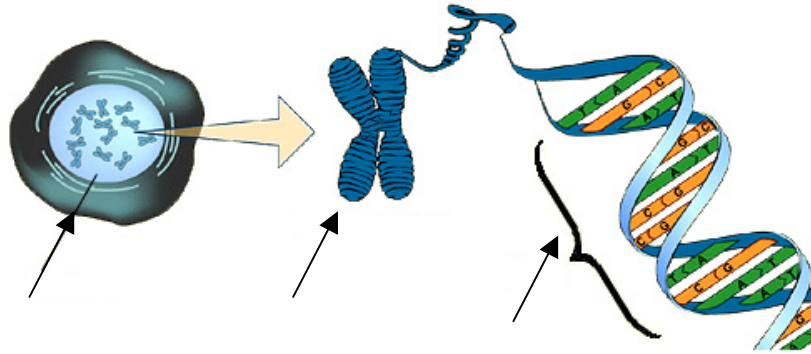




Genetic Variation

Label the following diagram



Humans have ___ pairs of chromosomes in every cell. They exist in pairs because you get ____ of them from your mother and half from your _____.

Egg cells and sperm cells are called gametes. Each gamete contains half of the parents 46 chromosomes, one of each pair.

A gene is a section of DNA which codes for a specific characteristic e.g. hair colour. Since we have two of each chromosome, we have two copies of each gene, these are called alleles. You get one from each parent.

You can be homozygous; both alleles are the same or heterozygous where each allele is different.

Some genes are dominant and some are recessive. A dominant gene always expressed if present and a recessive gene only shows if there are two copies.

Your phenotype is a description of the characteristics that are shown.

Your genotype is a description of the exact combination of genes you have.

Brown hair (B) is dominant over blonde (b). If my phenotype is brown, I have brown hair, my genotype could be Bb (heterozygous) or BB (homozygous).

Worked example.

1. If a blonde person mated with a homozygous brown haired person what would their off spring look like?

If one parent is blonde and blonde hair is RECESSIVE their genotype must be bb meaning they carry two copies of the blonde allele b.

The other parent is homozygous; this means he also has both alleles the same. His phenotype is brown hair, so his genotype is BB which means he carries two copies of the brown allele B.

This table shows the genotypes and phenotypes of each parent.

Parents Phenotype	Blonde		Brown	
Parents Genotype	bb		BB	
Gametes	b	b	B	B

If one allele must come from each parent then we can draw this punnet square to show the possible combinations.

		From Mother	
		b	b
From Father	B	Bb	Bb
	B	Bb	Bb

All the children are heterozygous brown haired.

Their phenotype is brown hair and genotype is Bb.

2. What would the offspring look like between two heterozygous brown haired people?

Parents Phenotype	Brown		Brown	
Parents Genotype	Bb		Bb	
Gametes	B	b	B	b

		From Mother	
		B	b
From Father	B	BB	Bb
	b	Bb	bb

We write the ratio of the genotypes as BB:2Bb:bb

We write the ratio of phenotypes as 3 Brown : 1Blonde

3. If a blonde person mated with a heterozygous brown haired person what would their off spring look like?

Parents Phenotype	Blonde		Brown	
Parents Genotype	bb		Bb	
Gametes	b	b	B	b

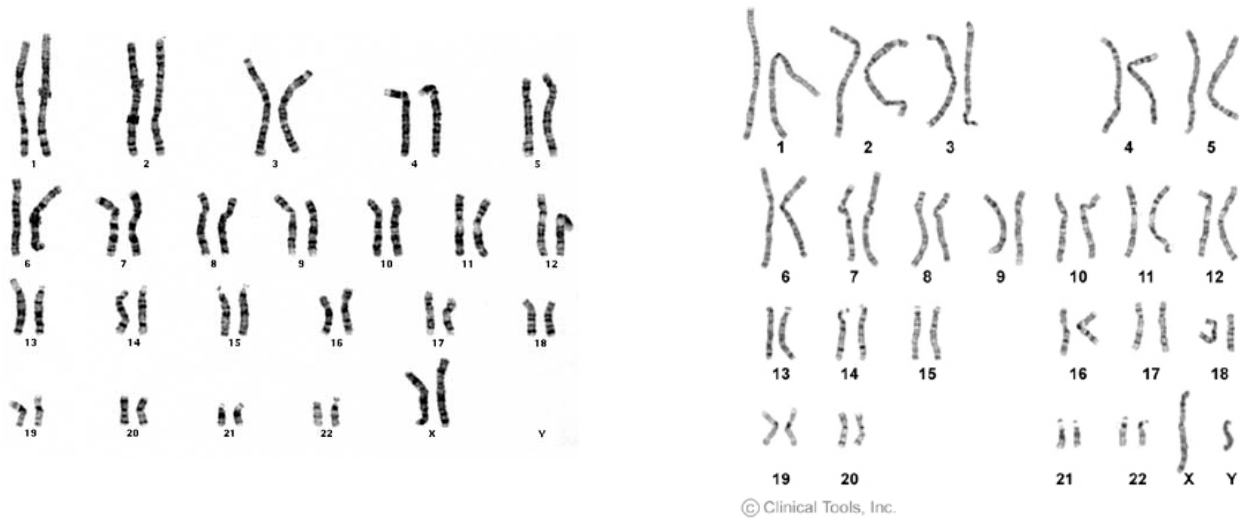
		From Mother	
		b	b
From Father	B	Bb	Bb
	b	bb	bb

This shows that there is an equal chance of the children having brown hair as blonde. We write the ratio of the genotypes as BB:bb

We write the ratio of phenotypes as 1 Brown : 1Blonde

Sex Determination.

The X and Y chromosomes determine your gender. Males have XY and females have XX. Label the diagram to show which sex these are.



Sex cells are haploid, they have half the number of chromosomes. Sex cells are also called gametes.

Father genotype = XY

Mother genotype = XX

This means that the father can give either an X or a Y chromosome and the mother can only give an X.

To show all the possible combinations we draw a punnet square.

		From Mother	
		X	X
From Father	Y	XY	XY
	X	XX	XX

This proves that there is always a 50:50 chance of getting either gender.

Genetic Variation Question Sheet

Information needed to answer these questions: (All dominant alleles are in capital letters)

Tall = T, Short = t
 Blonde hair = b, Brown hair = B
 Brown eyes = E, Blue eyes = e
 Red flowers = R, white flowers = r

1. Highlight any keywords in the text and write their definitions **in your book**.
2. Write out the genotype for the following organisms?

a. A HOMOZYGOUS red flower	
b. A HOMOZYGOUS tall plant	
c. A HETEROZYGOUS tall plant	
d. A HETEROZYGOUS Brown eyed person	
e. A HOMOZYGOUS blue eyed person.	

3. Using the genetic cross templates, work out the PHENOTYPE of the offspring of the following.

- a. A homozygous RED flower crossed with a homozygous WHITE flower.

Parents Phenotype		
Parents Genotype		
Gametes		

		From Mother	
From Father			

The genotypes are _____

The phenotypes are _____

- b. A heterozygous TALL plant crossed with a heterozygous TALL plant.

Parents Phenotype				
Parents Genotype				
Gametes				

		From Mother	
From Father			

The genotypes are _____

The phenotypes are _____

- c. Two blue eyed parents having children

Parents Phenotype				
Parents Genotype				
Gametes				

		From Mother	
From Father			

The genotypes are _____

The phenotypes are _____

d. Two heterozygous brown eyed parents having children.

Parents Phenotype				
Parents Genotype				
Gametes				

		From Mother	
From Father			

The genotypes are _____

The phenotypes are _____

Extension:

Explain, using genetic cross diagrams (punnet squares) why two brown eyed parents can have blue eyed offspring but two blue eyed parents can never have brown eyed offspring.

Genetic Variation Answer Sheet

Information needed to answer these questions: (All dominant alleles are in capital letters)

Tall = T, Short = t
Blonde hair = b, Brown hair = B
Brown eyes = E, Blue eyes = e
Red flowers = R, white flowers = r

1. Highlight any keywords in the text and write their definitions **in your book**.
2. Write out the genotype for the following organisms?

a. A HOMOZYGOUS red flower	RR
b. A HOMOZYGOUS tall plant	TT
c. A HETEROZYGOUS tall plant	Tt
d. A HETEROZYGOUS Brown eyed person	Bb
e. A HOMOZYGOUS blue eyed person.	bb

3. Using the genetic cross templates, work out the PHENOTYPE of the offspring of the following.

- a. A homozygous RED flower crossed with a homozygous WHITE flower.

Parents Phenotype	Red	White
Parents Genotype	RR	rr
Gametes	R R	r r

		From Mother	
		R	R
From Father	r	Rr	Rr
	r	Rr	Rr

The ratio of the genotypes: All Heterozygous Red.

The ratio of phenotypes: All Red flowers.

b. A heterozygous TALL plant crossed with a heterozygous TALL plant.

Parents Phenotype	Tall		Tall	
Parents Genotype	Tt		Tt	
Gametes	T	t	T	t

		From Mother	
		T	t
From Father	T	TT	Tt
	t	Tt	tt

The genotypes are _____ 3 Tall : 1 Short _____

The phenotypes are _____ 1 TT : 2 Tt : 1 tt _____

c. Two blue eyed parents having children

Parents Phenotype	Blue eye		Blue eyes	
Parents Genotype	bb		bb	
Gametes	b	b	b	b

		From Mother	
		b	b
From Father	b	b	b
	b	b	b

The phenotypes are _____ all blue eyes _____

The genotypes are _____ all homozygous blue = bb

d. Two heterozygous brown eyed parents having children.

Parents Phenotype	Brown		Brown	
Parents Genotype	Bb		Bb	
Gametes	B	b	B	b

		From Mother	
		B	b
From Father	B	BB	Bb
	b	Bb	bb

The phenotypes are _____ 3 brown eyed : 1 blue eyed _____

The genotypes are _____ 1 BB : 2 Bb : 1 bb

Extension:

Explain, using genetic cross diagrams (punnet squares) why two brown eyed parents can have blue eyed offspring but two blue eyed parents can never have brown eyed offspring.